Imperial College Business School

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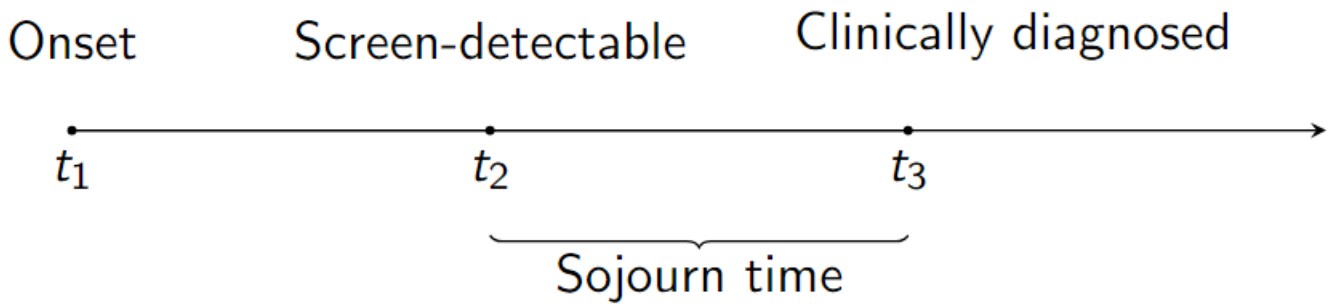
INTRODUCTION TO DATA SCIENCE

Tutorial 1

Autumn 2024

My research

- Multi-state survival models
  - Natural history of prostate cancer

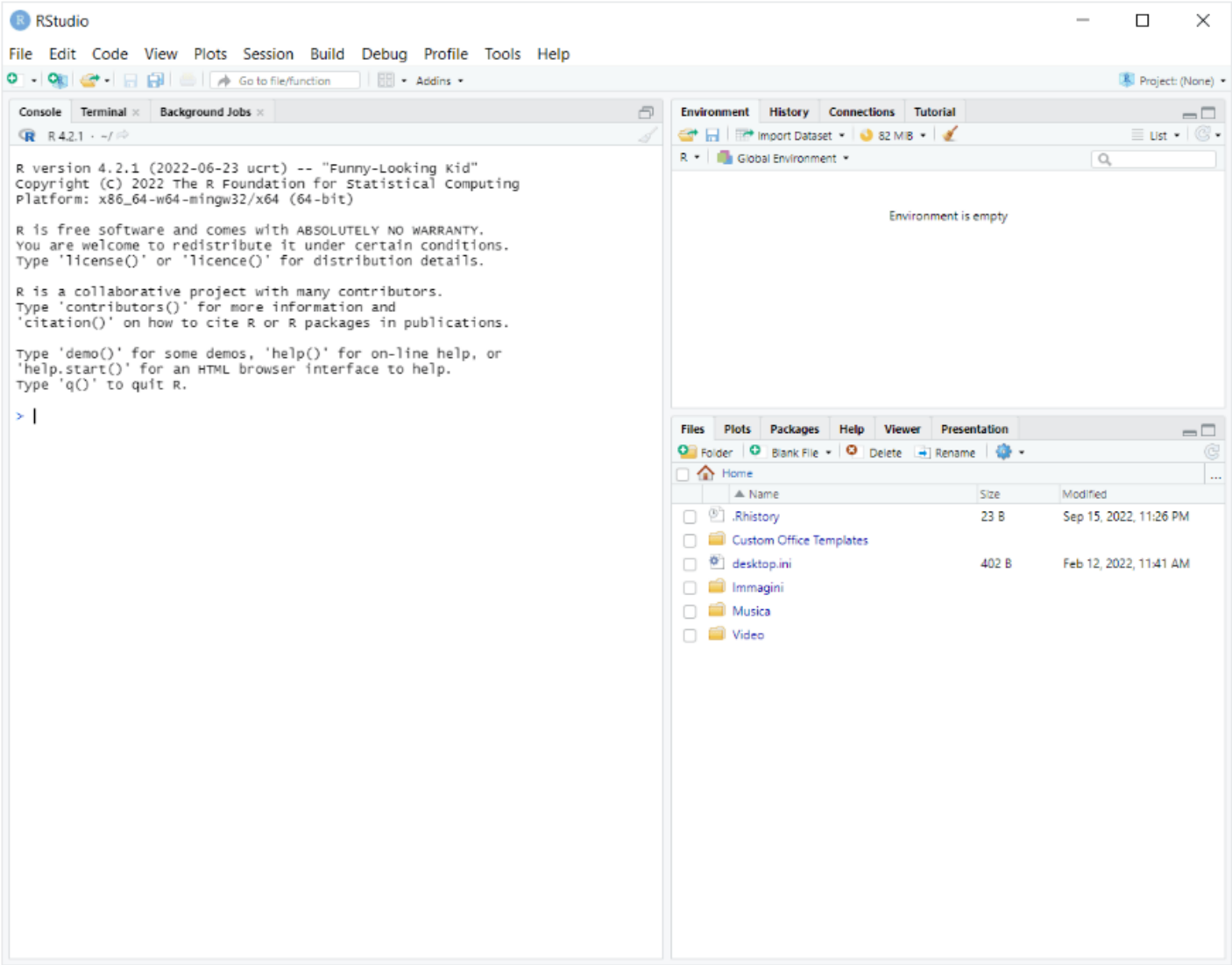


Screening is looking for cancer before a person has any symptoms.

- The prostate-specific antigen, is a blood test. The blood level of PSA is often elevated in people with prostate cancer.

R and Rstudio introduction

- A user-friendly interface
- Write and save reusable scripts
- Plot previewing
- Easy to switch between terminal and console
- Easy access to objects and imported data



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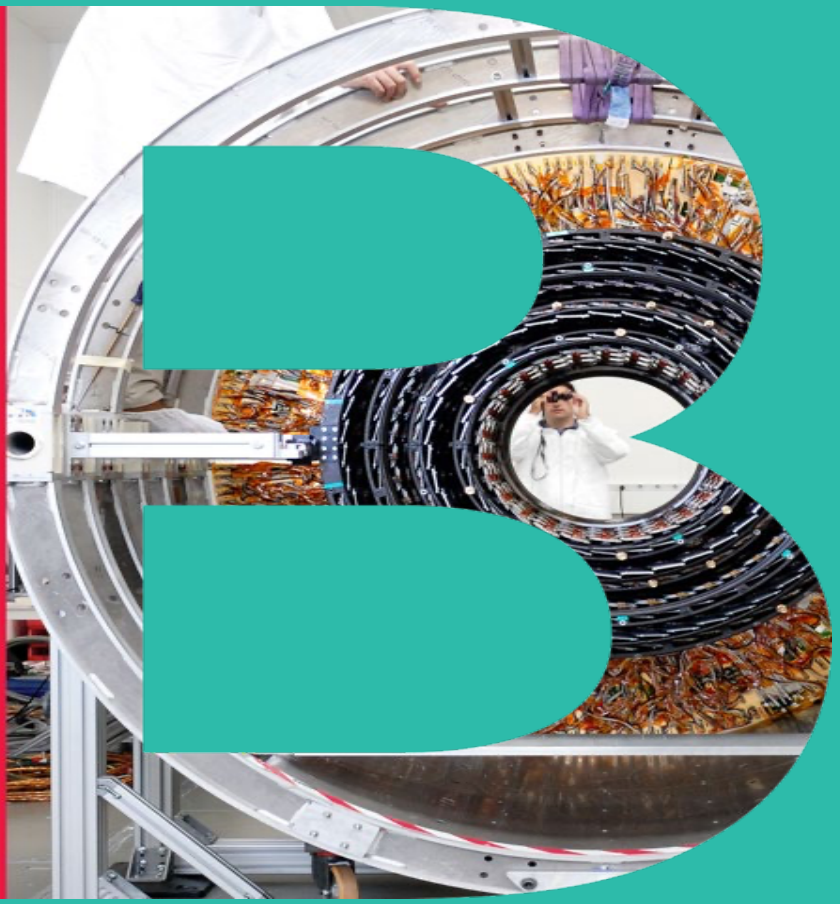

Tutorial #1

Introduction to Data Science

Data types in R

- **numeric -** (10.5, 55, 787)
- **integer -** (1L, 55L, 100L, where the letter "L" declares this as an integer)
- **complex -** (9 + 3i, where "i" is the imaginary part)
- **character** (a.k.a. string) - ("k", "R is exciting", "FALSE", "11.5")
- **logical** (a.k.a. boolean) - (TRUE or FALSE)
- **Date -** "07/01/1867“

- **Built-in functions in R.** A function in R is an object containing multiple command that are run together in a predefined order every time.

Best practices in coding. R

1. Set your working directory setwd()
2. Comment your code – helps you and others read the code
3. Libraries that are required at the beginning of the script
4. Name and style code consistently
5. Avoid naming scripts with spaces or long names
6. Use descriptive variable and function names
7. Keep all of the source files for a project in one directory and use relative paths to access them.